Global attractor and arbitrary-order Sobolev regularity for the axisymmetric Darcy–Bénard system

Detailed proof draft

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Scope of this draft. The phase space in the theorem below is the axisymmetric subspace of $H^1_0(Q)$, not all of $L^2(Q)$. The theorem establishes a global attractor and uniform bounds of every spatial Sobolev order on that attractor. It does not assert membership in every power domain of the homogeneous Dirichlet Laplacian. Fractional-power estimates are used for local well-posedness and compactness; boundary-aware parabolic and elliptic regularity are used for the final bootstrap. The physical boundary interpretation and the pressure normalization specified below are necessary clarifications of the supplied introduction.

1 Geometry, boundary conditions, and the pressure operator

Let $D_a = \{x' \in \mathbb{R}^2 : |x'| < a\}$ and $Q = D_a \times (0,1)$. The meridional domain is $\Omega = (0,a)\times(0,1)$ with measure $\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}z$. An axisymmetric scalar $f(r,z)$ is identified with $\widetilde{f}(x',z) = f(|x'|,z)$. Define

$$H^k_r(\Omega) = \{f : \widetilde{f} \in H^k(Q)\}, \qquad \|f\|_{H^k_r} = (2\pi)^{-1/2}\left\|\widetilde{f}\right\|_{H^k(Q)}.$$

This definition includes the regularity conditions at the symmetry axis. For $k \geq 2$ it is not to be replaced without proof by a sum containing only unweighted coordinate derivatives. Vector norms are defined through the corresponding Cartesian axisymmetric, swirl-free vector fields. Put

$$H = L^2_r(\Omega), \qquad V = \{f : \widetilde{f} \in H^1_0(Q),\ \widetilde{f} \text{ is axisymmetric}\}, \qquad \|f\|_V = \|\nabla_r f\|_{L^2_r}.$$

All constants below may depend on a and R, but not on an initial condition in a specified bounded set, unless explicitly indicated.

The equations are

$$-\nabla_r p - u + R\rho e_z = 0, \qquad \operatorname{div}_r u = 0, \qquad \rho_t + u\cdot\nabla_r\rho = Ru_z + \Delta_r\rho, \tag{1}$$

where

$$\operatorname{div}_r u = r^{-1}\partial_r(ru_r) + \partial_z u_z, \qquad \Delta_r = \partial_{rr} + r^{-1}\partial_r + \partial_{zz}.$$

The physical boundary conditions are

$$\rho = 0, \quad u_r = 0 \quad (r = a), \qquad \rho = 0, \quad u_z = 0 \quad (z = 0,1). \tag{2}$$

The axis $r = 0$ is not a physical Dirichlet boundary. For smooth fields, axis regularity includes $\partial_r\rho = \partial_r p = 0$ and $u_r = 0$ there, as well as the full smooth Cartesian extension conditions. The velocity condition in (**??**) is impermeability, not a homogeneous Dirichlet condition on all components of u.

Write $p = Rq$ and normalize q to have zero mean. For smooth states,

$$\Delta_r q = \partial_z\rho, \qquad \partial_n q = 0, \qquad u = R(\rho e_z - \nabla_r q).$$

For rough states the pressure is defined variationally, rather than by assigning a boundary trace to an arbitrary L^2 function. The Helmholtz projection P and $Kf = [P(fe_z)]_z$ give

$$\rho_t + A\rho = F(\rho), \qquad A = -\Delta_r, \quad \mathcal{U}(f) = RP(fe_z), \quad F(f) = -\mathcal{U}(f)\cdot\nabla_r f + R^2 Kf. \tag{3}$$

The L^2 orthogonal projection property gives

$$\|Kf\|_H \le \|f\|_H, \qquad (Kf, f)_H = \|P(fe_z)\|_H^2 \ge 0. \tag{4}$$

1.1 Reflection that removes the horizontal edges

Use the smooth product manifold with boundary

$$\mathcal{M} = D_a \times (\mathbb{R}/2\mathbb{Z}).$$

Its only boundary is $\partial D_a \times (\mathbb{R}/2\mathbb{Z})$. Extend ρ oddly from $(0,1)$ to $(-1,1)$ and then periodically with period 2. If $\rho \in V$, its extension belongs to $H^1(\mathcal{M})$, has zero lateral trace, and has comparable norm. No higher-order compatibility assumption on the initial condition is imposed by this H^1 extension.

For an extended scalar f, define q_f by

$$\Delta q_f = \partial_z f \quad \text{in } \mathcal{M}, \qquad \partial_n q_f = 0 \quad \text{on } \partial\mathcal{M}, \qquad \int_{\mathcal{M}} q_f = 0. \tag{5}$$

Periodicity makes the compatibility condition automatic. Equivalently,

$$\int_{\mathcal{M}} \nabla q_f \cdot \nabla\varphi = \int_{\mathcal{M}} f\partial_z\varphi.$$

The vector fe_z is tangent to the lateral boundary. Standard Neumann regularity, including its weak L^p form and interpolation, gives

$$\|\mathcal{U}(f)\|_{W^{s,p}(\mathcal{M})} + R^2\|Kf\|_{W^{s,p}(\mathcal{M})} \le C_{s,p,R}\|f\|_{W^{s,p}(\mathcal{M})}, \qquad s \ge 0, \quad 1 < p < \infty. \tag{6}$$

Here and below noninteger $W^{s,p}$ spaces have their usual Sobolev–Slobodeckij interpretation. At $s = 0$, one first uses the weak Neumann estimate for divergence-form data; at integer $s \ge 1$ one uses elliptic regularity, and intermediate indices follow by interpolation. The cylindrical L^p projection and reflection mechanism are also treated in [**?**]. We do not use boundedness of P on L^∞.

If f is odd in z, then q_f is even, by uniqueness in (**??**). Thus $\mathcal{U}(f)'$ is even and $\mathcal{U}(f)_z$ is odd. Consequently,

$$\mathcal{U}(f)'\cdot\nabla' f + \mathcal{U}(f)_z\partial_z f \quad \text{and} \quad Kf$$

are odd. The equation therefore preserves oddness and axisymmetry. Restricting a solution to $0 < z < 1$ gives $\rho = 0$ and $u_z = 0$ at the horizontal faces. This is an equivalent reflection construction, not a replacement of the original physical boundary conditions by periodic ones. It permits the use of smooth-boundary estimates on $\mathcal{M}$ without assuming arbitrary-order elliptic regularity on a domain with edges.

2 Main result and analytic tools

Let A also denote the positive Dirichlet Laplacian on the odd, axisymmetric scalar subspace of $L^2(\mathcal{M})$, with zero lateral trace. Its restriction to Q is the scalar Laplacian with the physical temperature boundary conditions. Let

$$X_\sigma = D(A^\sigma) \quad (\sigma \ge 0), \qquad X_{-\sigma} = (X_\sigma)',$$

where the dual is taken with H as pivot. In particular $X_0 = H$ and $X_{1/2} = V$. The harmless fixed normalization from reflection and the angular integration will be suppressed in estimates.

Theorem 2.1. *Fix $a > 0$ and $R > 0$. Interpret (??)–(??) as above and take $\rho_0 \in V$. Then:*

1. *There is a unique global mild solution in $C([0,\infty); V)$. It has the local energy regularity $L^2_{\mathrm{loc}}([0,\infty); D(A))$ and is smooth for positive time after the reflection described above. The solution maps form a continuous semiflow $S(t)$ on V.*

2. *$S(t)$ has a compact global attractor $\mathcal{A} \subset V$. For every bounded $B \subset V$,*

$$\sup_{b\in B} \inf_{a\in\mathcal{A}} \|S(t)b - a\|_V \longrightarrow 0.$$

3. *For every integer $k \geq 0$ there is $C_k = C_k(a, R)$ such that*

$$\sup_{\rho\in\mathcal{A}} \|\rho\|_{H^k_r(\Omega)} \leq C_k. \tag{7}$$

The attractor is compact in each of these Sobolev topologies. Every element has a smooth Cartesian axisymmetric extension up to the physical boundary. Velocity and normalized pressure have arbitrary spatial Sobolev regularity as well.

Remark 2.2. *The theorem does not assert a continuous semiflow on all of H. It also does not identify X_σ with the whole Sobolev space $H^{2\sigma}_r$. High power domains contain additional boundary conditions; see [?]. These distinctions are essential to the statement, not merely notational choices.*

We use the following standard linear facts, stated here to make all regularity inputs explicit. The analytic-semigroup estimates are those of [?, ?]; Dirichlet maximal regularity is covered by [?]. On the smooth periodic product $\mathcal{M}$, these follow by the same localization and boundary flattening arguments as on a smooth bounded domain.

1. For $d \geq c$,

$$\left\|e^{-tA}\right\|_{X_c\to X_d} \leq C_{c,d} t^{-(d-c)} e^{-\delta t}, \qquad t > 0. \tag{8}$$

Negative c is allowed by extrapolation. Also, $X_{1/4} \hookrightarrow L^3$ by interpolation and the three-dimensional Sobolev embedding, and $X_b \Subset X_c$ for $b > c$.

2. Put

$$\mathbb{E}_p(I) = W^{1,p}(I; L^p(\mathcal{M})) \cap L^p(I; W^{2,p}(\mathcal{M}) \cap W^{1,p}_0(\mathcal{M})).$$

The zero trace here is only on the lateral boundary. On intervals of fixed length, the zero-initial-data Dirichlet problem $w_t - \Delta w = h$ satisfies

$$\|w\|_{\mathbb{E}_p(I)} \leq C_p \|h\|_{L^p(I\times\mathcal{M})}. \tag{9}$$

One may apply this to $w = \chi(t)f(t)$, where χ vanishes near the left endpoint. Then the additional source is $\chi' f$.

3. The parabolic Sobolev and trace embeddings in spatial dimension three are

$$\|\nabla w\|_{L^{5p/(5-p)}(I\times\mathcal{M})} \leq C_p \|w\|_{\mathbb{E}_p(I)}, \qquad 1 < p < 5, \tag{10}$$

and, for $p = 10$,

$$\mathbb{E}_{10}(I) \hookrightarrow C(\bar{I}; W^{9/5,10}(\mathcal{M})), \qquad W^{9/5,10}(\mathcal{M}) \hookrightarrow W^{1,\infty}(\mathcal{M}). \tag{11}$$

The time trace is initially the Besov space $B^{9/5}_{10,10}$, which equals the noninteger Sobolev–Slobodeckij space used here.

4. Smooth-boundary Dirichlet elliptic regularity gives

$$\|w\|_{W^{s+1,10}} \leq C_s(\|Aw\|_{W^{s-1,10}} + \|w\|_{L^{10}}), \quad s > 1, \qquad w|_{\partial\mathcal{M}} = 0. \tag{12}$$

This is an estimate in ordinary Sobolev spaces; it does not require Aw to have zero trace.

3 Local well-posedness, dissipativity, and the attractor

3.1 Local well-posedness on V

By (**??**), Hölder's inequality, and $H^1 \hookrightarrow L^6$,

$$\|\mathcal{U}(f)\cdot\nabla f\|_{L^{3/2}} \le \|\mathcal{U}(f)\|_{L^6}\,\|\nabla f\|_{L^2} \le C_R\,\|f\|_V^2\,.$$

Since $X_{1/4} \hookrightarrow L^3$, duality gives $L^{3/2} \hookrightarrow X_{-1/4}$. Applying the same estimate to the difference of two products yields

$$\|F(f)-F(g)\|_{X_{-1/4}} \le C_R(1+\|f\|_V+\|g\|_V)\,\|f-g\|_V\,. \tag{13}$$

The mild equation is

$$\rho(t) = e^{-tA}\rho_0 + \int_0^t e^{-(t-s)A}F(\rho(s))\,\mathrm{d}s.$$

The kernel from $X_{-1/4}$ to $X_{1/2}$ has exponent $3/4$. On a ball of radius M in $C([0,T];V)$, the difference of the integral maps is bounded by

$$C_R(1+2M)\int_0^T s^{-3/4}\,\mathrm{d}s\,\|\rho-\eta\|_{C([0,T];V)} = 4C_R(1+2M)T^{1/4}\,\|\rho-\eta\|_{C([0,T];V)}\,.$$

For sufficiently small T this is a contraction and preserves the chosen ball. Thus there is a unique local solution, continuous dependence in V, and the continuation criterion

$$T_{\max}<\infty \implies \limsup_{t\uparrow T_{\max}}\|\rho(t)\|_V=\infty. \tag{14}$$

The energy tests below can first be performed for smooth approximations of the initial data. On a short common time interval, the H^1 estimate before using the maximum principle is $y'+\|A\rho\|_2^2 \le C_R(y^3+y+1)$, where $y=\|\rho\|_V^2$. It supplies uniform local $L^2(D(A))$ control. Passage to the mild solution uses its uniqueness and strong $C(V)$ convergence on shorter existence intervals. Truncation tests are justified by convex approximation. This provides the energy regularity and justifies the estimates independently of any attractor assertion.

3.2 Total-temperature truncations and a drift-independent bound

On the physical cylinder put

$$b(z)=R(1-z), \qquad \Theta=\rho+b, \qquad h_+=(\Theta-R)_+, \quad h_-=(-\Theta)_+.$$

Since $\nabla b=-Re_z$ and $\Delta b=0$,

$$\Theta_t+u\cdot\nabla\Theta=\Delta\Theta, \qquad \Theta|_{\partial Q}=b\in[0,R]. \tag{15}$$

Both truncations have zero trace. The divergence-free and impermeability properties cancel the transport term, so

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|h_\pm\|_2^2+\|\nabla h_\pm\|_2^2\le 0.$$

If $\lambda_1>0$ is the first physical Dirichlet eigenvalue and $E(t)^2=\|h_+(t)\|_2^2+\|h_-(t)\|_2^2$, then

$$E(t)\le e^{-\lambda_1(t-s)}E(s), \qquad E(0)\le\|\rho_0\|_2\,. \tag{16}$$

Here is the smoothing argument needed for initially nonbounded temperatures. Freeze the velocity of the local solution and denote the zero-Dirichlet convection–diffusion evolution by

$U(t,s)$. Positivity and testing with smooth approximations of the sign give $\|U(t,s)f\|_1 \le \|f\|_1$. The three-dimensional Nash inequality is

$$\|v\|_2^{10/3} \le C_N \|v\|_1^{4/3} \|\nabla v\|_2^2 .$$

Thus $Y(t) = \|U(t,s)f\|_2^2$ satisfies

$$Y' \le -\frac{2}{C_N \|f\|_1^{4/3}} Y^{5/3}.$$

Integration gives $\|U(t,s)\|_{L^1 \to L^2} \le C(t-s)^{-3/4}$. Apply the same argument to the backward adjoint, whose reversed drift is again divergence-free and tangent to the boundary, and use duality to obtain

$$\|U(t,s)\|_{L^2 \to L^\infty} \le C(t-s)^{-3/4}.$$

Each $h_\pm$ is a nonnegative subsolution, by convex approximation of the positive part. Positivity and comparison therefore give

$$\|h_\pm(t)\|_\infty \le C(t-s)^{-3/4} E(s), \qquad t > s. \tag{17}$$

The constants are independent of the size of the frozen drift.

Clipping Θ to $[0, R]$ shows

$$\|\rho(t)\|_2 \le R|Q|^{1/2} + E(t), \qquad \|\rho(t)\|_\infty \le R + \max_\pm \|h_\pm(t)\|_\infty .$$

For every bounded $B \subset V$, (**??**)–(**??**) with $s = t-1$ imply an entering time T_B such that

$$\|\rho(t)\|_2 \le M_0, \qquad \|\rho(t)\|_\infty \le M_\infty := R+1, \qquad t \ge T_B, \quad \rho_0 \in B. \tag{18}$$

One may take $M_0 = R|Q|^{1/2} + 1$ with the three-dimensional normalization, or the analogous weighted-volume constant.

3.3 Global continuation and the H^1 absorbing bound

Testing by ρ gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\rho\|_2^2 + \|\nabla\rho\|_2^2 = R^2(K\rho,\rho) \le R^2 \|\rho\|_2^2 . \tag{19}$$

For smooth solutions the equivalent pressure form has $+\|\nabla p\|_2^2$ on the left and $R^2\|\rho\|_2^2$ on the right.

Test by $A\rho$ and put $y = \|\nabla\rho\|_2^2$, $D = \|A\rho\|_2$. Then

$$\tfrac{1}{2}y' + D^2 = R^2(K\rho, A\rho) - (u\cdot\nabla\rho, A\rho).$$

The linear term satisfies

$$|R^2(K\rho, A\rho)| \le \tfrac{1}{4}D^2 + R^4\|\rho\|_2^2 .$$

The three-dimensional interpolation and elliptic estimates give

$$\|\nabla\rho\|_3 \le C\|\nabla\rho\|_2^{1/2}\|A\rho\|_2^{1/2} .$$

By (**??**) and (**??**), $\|u\|_6 \le C_R\|\rho\|_6 \le U_6$ for $t \ge T_B$. Consequently,

$$\begin{aligned} |(u\cdot\nabla\rho, A\rho)| &\le \|u\|_6\|\nabla\rho\|_3 D \le CU_6 y^{1/4} D^{3/2} \\ &\le \tfrac{1}{4}D^2 + CU_6^4 y. \end{aligned}$$

Hence, with fixed $a_0, b_0 \geq 0$,

$$y' + D^2 \leq a_0 y + b_0, \qquad t \geq T_B. \tag{20}$$

This also proves global continuation before an absorbing time is available. If $T_{\max} < \infty$, fix $0 < \tau < T_{\max}$. Equations (**??**)–(**??**) give a finite bound for $\|\rho\|_\infty$ on $[\tau, T_{\max})$. The same computation therefore yields $y' \leq C_\tau y + C_\tau$. Ordinary Gronwall bounds y up to $T_{\max}$, contradicting (**??**). The solution is global.

Integrating (**??**) over $[t, t+1]$, now for $t \geq T_B$, gives

$$\int_t^{t+1} y(s)\,\mathrm{d}s \leq (\tfrac{1}{2} + R^2) M_0^2 =: J. \tag{21}$$

The uniform Gronwall lemma applied to (**??**) gives

$$y(t) \leq M_1^2 := (J + b_0) e^{a_0}, \qquad t \geq T_B + 1. \tag{22}$$

Integration of (**??**) also gives

$$\int_t^{t+1} \|A\rho(s)\|_2^2 \,\mathrm{d}s \leq M_1^2 + a_0 J + b_0, \qquad t \geq T_B + 1. \tag{23}$$

Continuous dependence on any finite interval follows by repeating the local fixed-point argument along the global solution. Thus $S(t)$ is a continuous semiflow on V.

3.4 Compactness without an invalid $F : V \to H$ estimate

Equation (**??**) and (**??**) imply $\|F(\rho(t))\|_{X_{-1/4}} \leq C$ after absorption. For $t \geq T_B + 2$, use a fixed-length variation-of-constants formula:

$$\rho(t) = e^{-A}\rho(t-1) + \int_{t-1}^{t} e^{-(t-s)A} F(\rho(s))\,\mathrm{d}s.$$

By (**??**),

$$\|\rho(t)\|_{X_{2/3}} \leq C M_1 + C \int_0^1 \tau^{-11/12}\,\mathrm{d}\tau \leq C_*. \tag{24}$$

The exponent is $2/3 - (-1/4) = 11/12 < 1$. The V ball in (**??**) absorbs every bounded set of V, and (**??**), together with $X_{2/3} \Subset V$, proves eventual uniform compactness. The global-attractor theorem [**?**] therefore gives

$$\mathcal{A} = \omega_V(B_V).$$

This proves the first two assertions concerning global existence and the attractor. Positive-time smoothness will follow from the argument in the next section; that argument applies on finite windows bounded away from $t = 0$, as well as uniformly on the attractor.

4 Arbitrary-order Sobolev regularity

4.1 Complete trajectories and uniform low-order bounds

Every point of a compact invariant attractor lies on a complete trajectory $\rho : \mathbb{R} \to \mathcal{A}$. This follows by choosing successive preimages under $S(1)$, using compactness, and defining the solution on the intervening intervals. Invariance and the absorbing bounds imply

$$\sup_{t \in \mathbb{R}} \big(\|\rho(t)\|_V + \|\rho(t)\|_\infty \big) \leq C(a, R). \tag{25}$$

In fact (**??**) applied from $t-L$ to t gives $E(t) \le Ce^{-\lambda_1 L}$ for a complete bounded trajectory. Letting $L \to \infty$ yields $0 \le \Theta \le R$ and hence $\|\rho\|_\infty \le R$ on the attractor.

Reflect this trajectory onto $\mathcal{M}$. By (**??**),

$$\sup_{t\in\mathbb{R}} \|u(t)\|_{L^{20}(\mathcal{M})} \le C(a,R). \tag{26}$$

All subsequent estimates on windows of fixed length are uniform both in the window position and in the chosen complete trajectory.

4.2 Four explicit parabolic integrability improvements

Let $I_0 = (t-5, t+1)$ and choose nested windows $I_j = (t-5+j, t+1)$, $j = 1,2,3,4$. At each step choose a smooth time cutoff supported after the left endpoint of I_{j-1} and equal to one on I_j. The cutoff source $\chi'\rho$ belongs to every finite L^p with a uniform bound by (**??**). The equation for the cutoff is

$$(\chi\rho)_t - \Delta(\chi\rho) = -\chi u\cdot\nabla\rho + \chi R u_z + \chi'\rho.$$

Initially $\nabla\rho \in L^2(I_0\times\mathcal{M})$ and $u \in L^{20}$. Thus Hölder gives $F(\rho) \in L^{20/11}$, because $11/20 = 1/20 + 1/2$. Apply (**??**), then (**??**), and repeat on the smaller windows:

step	$F(\rho)$ in L^q	ρ in $\mathbb{E}_q$	$\nabla\rho$ in L^r
1	$q = 20/11$	$\mathbb{E}_{20/11}(I_1)$	$r = 20/7$
2	$q = 5/2$	$\mathbb{E}_{5/2}(I_2)$	$r = 5$
3	$q = 4$	$\mathbb{E}_4(I_3)$	$r = 20$
4	$q = 10$	$\mathbb{E}_{10}(I_4)$	bounded, by the trace embedding

The next forcing exponent always satisfies $1/q = 1/20 + 1/r$. All linear terms are in the required L^q, because $u \in L^{20}$ and the windows have bounded measure. Consequently,

$$\sup_{t\in\mathbb{R}} \|\rho\|_{\mathbb{E}_{10}((t,t+1))} \le C, \qquad \sup_{t\in\mathbb{R}} \|\rho(t)\|_{W^{9/5,10}(\mathcal{M})} \le C. \tag{27}$$

In particular

$$\sup_{t\in\mathbb{R}} (\|u(t)\|_\infty + \|\nabla\rho(t)\|_\infty) \le C. \tag{28}$$

The bound on u follows from (**??**) in $W^{9/5,10}$, not from an L^∞ projection estimate.

4.3 All time derivatives, with uniform window bounds

Write $v_j = \partial_t^j \rho$ and $v_0 = \rho$. We prove, by induction on j, that

$$\sup_{t\in\mathbb{R}} \|v_j\|_{\mathbb{E}_{10}((t,t+1))} \le C_j, \qquad \sup_{t\in\mathbb{R}} \|v_j(t)\|_{W^{9/5,10}} \le C_j. \tag{29}$$

The case $j = 0$ is (**??**). The operator $\mathcal{U}$ is time-independent and linear, hence commutes with time difference quotients and derivatives. Formal differentiation gives

$$(v_j)_t + Av_j = R^2 K v_j - \sum_{\ell=0}^{j} \binom{j}{\ell} \mathcal{U}(v_\ell)\cdot\nabla v_{j-\ell}. \tag{30}$$

The differentiated temperature has zero lateral trace, justified first for difference quotients and then by passage to the limit.

For clarity, the regularity step underlying this differentiation is given explicitly. If w solves

$$w_t + Aw + u\cdot\nabla w + \mathcal{U}(w)\cdot\nabla\rho - R^2 K w = h,$$

then (**??**), applied after a time cutoff, gives

$$\|\chi w\|_{\mathbb{E}_{10}} \le C\big(\|u\|_\infty \|\nabla(\chi w)\|_{10} + \|\nabla\rho\|_\infty \|\mathcal{U}(\chi w)\|_{10} + \|K(\chi w)\|_{10} + \|\chi h\|_{10} + \|\chi' w\|_{10}\big).$$

Use (**??**) in L^{10} and

$$\|\nabla z\|_{10} \le \varepsilon \left\|D^2 z\right\|_{10} + C_\varepsilon \|z\|_{10}.$$

By (**??**), ε can be chosen to absorb the second-derivative term. Thus on strictly nested fixed-length windows $J' \Subset J$ (with time separation only at their left endpoints),

$$\|w\|_{\mathbb{E}_{10}(J')} \le C\big(\|w\|_{L^{10}(J\times\mathcal{M})} + \|h\|_{L^{10}(J\times\mathcal{M})}\big). \tag{31}$$

The constant is uniform on the attractor.

Assume (**??**) is known through j. Since $v_j \in W^{1,10}(J; L^{10})$, its difference quotients $w_h = [v_j(\cdot + h) - v_j]/h$ are uniformly bounded in L^{10} on a shorter window. Subtracting the equations for v_j gives the linearized equation above, with shifted versions of the same bounded coefficients. Its remaining source consists of products of $\mathcal{U}(v_\ell)$ and ∇v_m with $\ell, m \le j$, or their time averages. These are uniformly bounded in L^{10}: (**??**), (**??**), and the induction hypothesis bound the velocity factor and the spatial-gradient factor in L^∞. Applying (**??**) gives a uniform $\mathbb{E}_{10}$ bound for w_h. Weak compactness and the convergence of difference quotients in distributions give $v_{j+1} \in \mathbb{E}_{10}$. The trace theorem gives its stated spatial bound.

Equivalently, after passage to the limit, the nonprincipal source in the equation for v_{j+1} is

$$-\sum_{\ell=1}^{j} \binom{j+1}{\ell} \mathcal{U}(v_\ell) \cdot \nabla v_{j+1-\ell},$$

whose indices are all at most j. This verifies that the induction does not use an unproved bound for v_{j+1}. All finite losses of time-window length are harmless for complete trajectories: begin with a longer window. This proves (**??**) for every j.

4.4 Spatial induction in ordinary Sobolev spaces

Let $s_0 = 9/5$. We now prove that for every $m, j \ge 0$,

$$\sup_{t\in\mathbb{R}} \|v_j(t)\|_{W^{s_0+m,10}(\mathcal{M})} \le C_{j,m}. \tag{32}$$

The case $m = 0$ is (**??**). Suppose it holds for all j at a fixed m, and put $s = s_0 + m$. Rearranging (**??**),

$$Av_j = -v_{j+1} + R^2 K v_j - \sum_{\ell=0}^{j} \binom{j}{\ell} \mathcal{U}(v_\ell) \cdot \nabla v_{j-\ell}.$$

Since $s - 1 \ge 4/5 > 3/10$, $W^{s-1,10}(\mathcal{M})$ is a Banach algebra. The multiplication estimate, in the applicable range of [**?**], and (**??**) imply

$$\begin{aligned}\|\mathcal{U}(v_\ell) \cdot \nabla v_{j-\ell}\|_{W^{s-1,10}} &\le C_s \|\mathcal{U}(v_\ell)\|_{W^{s-1,10}} \|\nabla v_{j-\ell}\|_{W^{s-1,10}} \\ &\le C_{s,R} \|v_\ell\|_{W^{s,10}} \|v_{j-\ell}\|_{W^{s,10}}.\end{aligned}$$

The right-hand side of the elliptic equation is therefore uniformly bounded in $W^{s-1,10}$. Equation (**??**) gives

$$\sup_t \|v_j(t)\|_{W^{s+1,10}} \le C_{j,m+1}.$$

For a prescribed finite j, m, only finitely many time-derivative bounds are needed (up to a finite larger index). Thus the two-index induction contains no circular use of infinite regularity.

Taking $j = 0$ and choosing m large enough, the bounded-domain Sobolev inclusions give

$$\sup_t \|\rho(t)\|_{H^k(\mathcal{M})} \le C_k \qquad (k \ge 0).$$

Restriction to Q and the definition of H^k_r prove (**??**), uniformly over all complete trajectories and therefore over all of $\mathcal{A}$. Applying (**??**) and (**??**) gives the corresponding velocity and pressure regularity. Sobolev embedding for arbitrarily large orders gives smoothness up to the boundary of $\mathcal{M}$, hence up to the physical faces and their intersections after restriction.

Finally, uniform bounds in $H^{k+1}(Q)$ and compactness in V imply compactness of $\mathcal{A}$ in $H^k(Q)$ by interpolation and Rellich compactness. The same local-in-time bootstrap, beginning on intervals strictly after zero, proves the positive-time smoothness stated in Theorem **??**. For an arbitrary finite order the required number of nested intervals is finite; they can be chosen inside any given positive-time interval, with constants depending on its distance from $t = 0$. This completes the proof. □

5 Consequences for the manuscript formulation

The proof establishes the requested existence and arbitrary-order *Sobolev* regularity with phase space V. To claim an H-global attractor for a continuous semiflow on all H, a separate construction, uniqueness theorem, and continuous-dependence argument for initial data in H are still required. They are not supplied by the V result.

The scalar diffusion remains second order. Its fractional spaces have Sobolev order 2σ in the ranges where the proper boundary conditions are included, not order 4σ. The LZ iterative idea [**?**] is useful for organizing smoothing estimates, but its fourth-order operator and its nonlinear mapping properties cannot be transferred unchanged.

For a smooth solution, on the lateral wall the temperature equation and impermeability give

$$A\rho = Ru_z.$$

There is no boundary assumption $u_z = 0$ on that wall. Thus one cannot infer the trace condition $A\rho = 0$ required for membership in $D(A^2)$. This does not prevent the ordinary Sobolev regularity proved above: elliptic regularity with homogeneous boundary data for ρ does not require homogeneous boundary data for its forcing.

References

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